

Abel Zelalem

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Education

Bachelor of Science: Computer Engineering
University of Nevada, Las Vegas (UNLV)

Expected May 2027

Associate of Science
College of Southern Nevada

May 2024

Technical Skills

Languages: C | C++ | SystemVerilog | Verilog | RAPID | Python | Java (Familiar)

Architectures: AVR (ATmega328P/PB, ATmega2560) | ARM | RISC-V | MIPS

Hardware and Circuits: Microcontrollers | Digital Logic | Analog Circuits | PCB Design (KiCad) | Oscilloscope | LTspice

Robotics & CAD: ABB Robotstudio | RAPID | Pick-and-Place Automation | Fusion 360 | 3D Printing

Applications: Git | Microchip Studio | ModelSim | Quartus II | MATLAB | VS Code

Experience

Robotics Programmer

February 2026 - Present

Naama Hair and Wigs LLC

Las Vegas, NV

- Programmed the ABB YuMi IRB 14000 dual-arm collaborative robot in RAPID and RobotStudio to perform pick-and-place sequences for wig assembly automation, building the full motion control pipeline from scratch with no prior internal robotics infrastructure
- Designed and 3D printed custom process fixturing jigs in Fusion 360 to constrain hair strand positioning within the robot workspace, enabling repeatable manipulation of flexible materials that previously required fully manual assembly.
- Soole robotics engineer at an early-stage startup — independently scoped, prototyped, and iterated on hardware and software with no existing automation baseline, producing working prototype demonstrations of targeted assembly tasks.

Store Associate

November 2021 - November 2023

In-N-Out Burger

Las Vegas, NV

- Worked at one of the highest-volume In-N-Out locations in the country, consistently meeting speed and quality targets during peak rushes while coordinating with a team of 15–20 people.
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Project Experience

Autonomous Obstacle-Avoiding Rover, Personal Project

Summer 2026

- Building a 4-wheel autonomous rover on the ATmega2560 (Arduino Mega) with HC-SR04 ultrasonic and IR sensors for obstacle detection, implementing PID motor speed control loops using encoder pulse counting on external interrupt pins.
- Transmitting live telemetry (RPM, sensor distance, state) over Bluetooth via HC-05 UART to a phone terminal; designing a custom motor driver shield in KiCad as a PCB stretch goal.

Embedded Multi-Sensor Data Logger, Personal Project

Summer 2026

- Building an environmental monitoring station on the ATmega2560 that logs temperature, humidity, barometric pressure, and air quality readings to a microSD card in timestamped CSV format using a DS3231 RTC.
- Firmware uses a cooperative task scheduler written from scratch — seven concurrent tasks dispatched by elapsed-time comparison with zero blocking delay() calls, demonstrating real-time scheduling concepts without an RTOS.

SystemVerilog Elevator, Digital Logic Design II

April 2025

- Designed and simulated a multi-floor elevator controller in SystemVerilog, implementing floor request queuing, directional movement logic, and door sequencing within a structured finite state machine.
- Verified timing and functional correctness in ModelSim across edge-case scenarios including simultaneous multi-floor requests and door fault conditions.

Pulse Width Modulator (PWM) Circuit, Circuits II

August 2025

- Designed a variable duty cycle PWM circuit using a 555 timer and passive components; validated frequency and duty cycle performance against LTspice simulations using oscilloscope measurements.